

FHESore CSD

FPGA computational storage

Device Runtime

Submit task

Configure / execute

Storage-side data path

Persistent data
& results

Preprocess
(optional)

Encode & Encrypt
FPGA

Write-back

Decrypt & Decode
FPGA

PCIe

Host

Application

Task Runtime

Operator graph

Filter

Encrypt

Operator contexts

Data ranges

Network

Ciphertext relay

Input ciphertext relay

Result ciphertext relay

**Remote FHE
Service**

HPU

Homomorphic
evaluation

→ Data flow

- - -> Task control